



Development of stable MoO₃/TiO₂-Al₂O₃ catalyst for oxidative dehydrogenation of ethylbenzene to styrene using CO₂ as soft oxidant



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ABSTRACT

A novel MoO₃/TiO₂-Al₂O₃ (MTA) catalyst was employed for oxidative dehydrogenation of ethylbenzene to styrene using CO₂ as a soft oxidant with CO₂/ethylbenzene (molar ratio)=9.8 and gas hourly space velocity (GHSV) was maintained at 0.08 molh⁻¹ g⁻¹ catalyst. Among different (2.5, 5.0, 7.5 and 10.0) wt.% loadings of MoO₃, 7.5 wt.% loaded MoO₃/TiO₂-Al₂O₃ catalyst exhibited superior activity. The stability of the catalyst over time on stream was noteworthy where; the catalyst evinced notable activity up to 40 h with maximum ethylbenzene conversion of about 70% and styrene selectivity about 97%. The presence of titania greatly influenced the activity of the catalyst in terms of stability. The reaction with CO₂ resulted in better activity compared to N₂ proving it to be an effective oxidant for the dehydrogenation reactions. The deactivation of the catalyst was due to excess coke formation which led to blocking of active sites. The coke was removed by regenerating the catalyst at 600 °C under air. Interestingly, the catalysts were found active even after 40 h as well as in between 10 h of regenerations. The catalysts were characterized by X-Ray Diffraction, X-ray Photoelectron Spectroscopy, Surface area and Pore Size Distribution, Scanning Electron Microscopy, Thermogravimetric Analysis, H₂-Temperature Programmed Reduction and Ammonia-Temperature Programmed Desorption. The results indicate that the surface area, interaction with the support, morphology of the catalyst and surface acidic sites exerted a significant influence on the activity.

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1. Introduction

Styrene is one of the basic substrates for plastic industry, used mainly for the production of widely applied polystyrene and styrene copolymers such as acrylonitrile-butadiene-styrene (ABS), styrene acrylonitrile (SAN), styrene-butadiene rubber (SBR) etc. In past decade, extensive efforts have been made in the field of dehydrogenation of ethylbenzene to produce styrene owing to its enormous industrial application [1–4]. The conventional catalytic dehydrogenation method makes use of super-heated steam leading to a huge amount of energy consumption (1.5 × 10⁶ kcal/styrene ton). Intensive studies suggest that oxidative dehydrogenation (ODH) of ethylbenzene is an attractive substitute to catalytic dehydrogenation [5–7]. The method has potential to overcome drawbacks of the conventional method as it has higher thermodynamic stability [8–10]. Alternatively, it has been realized that

carbon dioxide (CO₂) utilization as a mild oxidant has various advantages such as acceleration of the reaction rate, enhancement in product selectivity, diminishing of thermodynamic limitations, suppression of total oxidation, prolonging of catalyst life, and so on. Also, CO₂ being inexpensive, nontoxic, renewable feedstock provides significant positive impact on the global carbon balance [3,11–13]. The proposed mechanism is that CO₂ abstracts hydrogen from simple or functionalized hydrocarbons through the catalytic activation to form CO and oxygen species [14–16]. In our recent review article, various widely studied catalysts for the ODH of ethylbenzene were reported [17]. Particularly, vanadium, iron, mixed-oxides and carbon based catalyst systems are found to be highly active for the reaction. There is strong evidence, that the actual catalytic surface in many reactions may be primarily carbon and carbonaceous substances due to which various carbon based catalysts have been studied recently [18–20]. Nevertheless, the mixed metal-oxide based catalyst has attracted considerable attention and the possibilities of TiO₂-ZrO₂, MnO₂-ZrO₂, V₂O₅-CeO₂, SnO₂-ZrO₂, FeO_x-Al₂O₃, TiO_x-Al₂O₃ and CeO₂-ZrO₂ as an acid-base bi-functional support as well as catalyst due to the increase in

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